Total: **/143**

[Dynamic Planet]

Troy 2021-2022 Invitational

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ANSWER KEY

Grading Notes:

- wording doesn't have to be exact
- accept free-response answers that don't match the answer key **IF** it makes sense scientifically
- for the winner of a tie, add 0.01 points to their score
- 143 points total

Tiebreaker Questions: 57, 72, 34 45 (in that specific order)

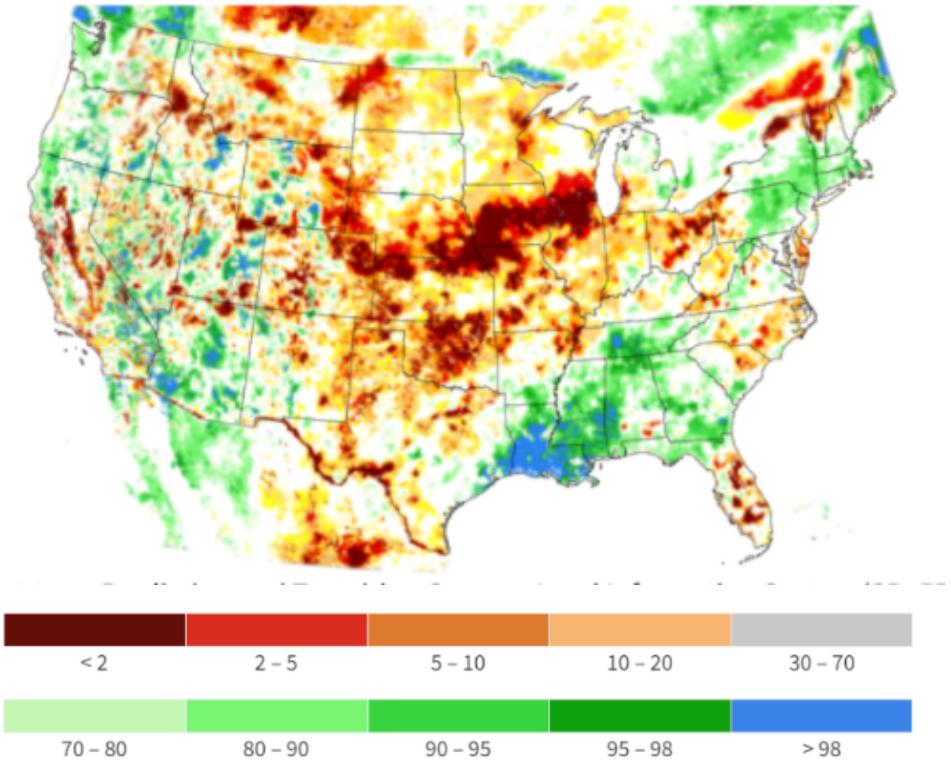
Part I: Multiple Choice (32 points)

1. Which of the following qualifies as condensation nuclei?
    a. Water droplet less than 1 micron
    b. Water droplet greater than 1 micron
    c. Water droplet equal to 1 micron
    d. Dust particles less than 1 micron
    e. Dust particle equal to or greater than 1 micron
2. What part of the floodplain is there less water, slower flow, and more friction?
    a. Inner bank of a meander
    b. Outer bank of a meander
    c. Meander scars
    d. Channel cut off
    e. Oxbow lake
    f. Two of the above
3. What is the opposite of channel degradation?
    a. Channel gradation
    b. Channel degradation
    c. Channel aggravation
    d. Channel deposition
    e. Channel erosion
4. Intersection of multiple alluvial fans are called--
    a. Channel
    b. Alluvial road
    c. Bajadas
    d. Valley
    e. Fjord
5. _____ is when water precipitates onto the soil, while _____ is when the water continues to seep through layers underneath the soil. Both of these processes [can or cannot] be used interchangeably.
    a. Penetration...infiltration...can
    b. Penetration...infiltration...cannot
    c. Infiltration...percolation...can
    d. Infiltration...percolation...cannot
    e. Infiltration...penetration...cannot
6. Soil with crusted edges leading to low infiltration rates are called _____.
    a. Outdated soil
    b. Impermeable soil
    c. Semi-permeable soil
    d. Hydrophobic soil

e. Hydrophilic soil

7. For a long time, groundwater flow was known for certain that it contributed to the _____ aspect of a storm hydrograph. However, it was not until the 1960s, when _____ proposed that groundwater might contribute to storm flow as well.
    a. Baseline; Robert Horton
    b. Baseline; Sklash and Farvolden
    c. Peak; Sklash and Farvolden
    d. Peak; Robert Horton
    e. Recession limb; Betson
8. Spheroidal weathering is a type of _____.
    a. Onion skin weathering
    b. Physical weathering
    c. Chemical weathering
    d. Biological weathering
    e. More than two of the above
9. Sediments follow what type of classification?
    a. Udden-Wentworth
    b. Hantel-Mentzel
    c. Hubbert-Horton
    d. Darwin-Linnaeus
    e. None of the above
10. Which of the following is NOT a defining characteristic of a rejuvenated valley/channel?
    a. Broad, flat plains
    b. Meanders
    c. Sandbars
    d. Rapids
    e. All of the above
    f. Answers b and d
11. Which of the following are possible effects of over extraction of wells?
    a. Subsidence
    b. Salt-water contamination
    c. Decreased sea levels
    d. Increased faulting
    e. Answers a and b only
    f. All of the above
12. An old river would likely have:
    a. Less energy dedicated to its load
    b. More energy dedicated to its load
    c. Balanced distribution of energy in its sediment load and for streamflow
    d. None of the above

13. A soil moisture content of 82 is optimal at:
    a. Fine clay w/ irrigation
    b. Fine clay without irrigation
    c. Medium clay with irrigation
    d. Medium clay without irrigation
    e. None of the above
    f. Answers b and c only
14. Root zone soil moisture applies to the upper [blank] meters below the surface
    a. There is no such thing as root zone soil moisture, soil measure only applies to the surface and atmosphere
    b. 100
    c. 200
    d. 225
    e. 250
15. What percentage of precipitation actually flows on land as runoff? Where do the rest go?
    a. 33%; groundwater and evaporation
    b. 33%; freezes to ice and sublimation
    c. 66%; groundwater, evaporation, and condensation
    d. 66%; freezes to ice and sublimation
16. What do you think the chart below best represents?



    a. Temperature
    b. Water availability

c. Soil-moisture
d. Air pressure

17. Which of the following are the main mechanisms in the formation of geysers? Select the best answer.
    a. Temperature and boiling point
    b. Hole size
    c. Pressure
    d. Salinity
    e. Acidity
18. Unstable channels found along fault lines are called-
    a. Straight
    b. Reactive
    c. Young
    d. Mature
    e. More than one of the above
19. True/False: Channels, rivers, and streams essentially mean the same thing, which is why they are used interchangeably.
    a. True
    b. False
20. Other than sinuosity, how can one determine what type of stream channel is present?
    a. Hydraulic gradient
    b. Sediment load
    c. Temperature
    d. Presence of eyots
    e. More than one of the above
21. Which of the following landforms, sediments, or processes are responsible for the fertile land by the Yellow RIver in China? Select the best answer.
    a. Floodplains & meanders
    b. Rich silt
    c. Loess
    d. Increased bioerosion, but decreased instances of aeolian transport
    e. None of the above
22. True/False: There are 3 stages to a meandering stream channel.
    a. True
    b. False
23. True/False: When the sinuosity of a channel is greater than three, it can be classified as an anastomosing channel.
    a. True
    b. False
24. True/False: Honey seeping into rocks can stimulate disaggregation.

a. True
b. False

25. True/False: When sediments are being collected, they can only be bedded or suspended.
    a. True
    b. False
26. True/False: The Sun is the main source of energy fueling the hydrologic cycle.
    a. True
    b. False
27. True/False: Though running water is a major mechanism shaping Earth's surface, it is not necessarily the most important.
    a. True
    b. False
28. As discharge increases, the sediment capacity-
    a. Decreases
    b. Increases
    c. Stays the same
    d. Not enough information
29. As stream velocity increases, how does that influence the stream competence?
    a. Increases
    b. Decreases
    c. Stays the same
    d. Not enough information
30. Given that the velocity of a stream is 4 meters squared per second, what happens to the stream competency?
    a. Stream competency increases 8x
    b. Stream competency decreases 8x
    c. Stream competency increases 16x
    d. Stream competency decreases 16x
    e. Not enough information
31. True/False: Soils that arise from the same parent soil likely have the same properties, and when looking at two soil samples that arise from two different types of parent soil, they likely have different properties.
    a. True
    b. False
32. True/False: In sheet erosion, ice is the leading intermediate that carries soil.
    a. True
    b. False

Part II: Short Answer (111 pts.)

33. (3 pts.) What are 3 conditions needed to achieve the state of precipitation during the water cycle?
    There must be some form of atmospheric cooling because the colder it is, less water molecules will be in its gaseous state, there must be condensation on nuclei, and the growth and accumulation of water droplets.
34. (3 pts.) How does the bergeron process and collision and coalescence correlate with water droplets? **(Tiebreaker #3)**
    They are responsible for water droplet growth in the clouds to at least 3,000 microns to begin precipitating because otherwise, the pull upward from the clouds exceeds the force exerted by gravity. Collision and coalescence is a relatively simple process of water droplets colliding and fusing with one another, and the Bergeron process accounts for water vapor pressure and influencing water droplet growth.
35. (2pts.) How might rain gauges underestimate rainfall?
    Rain gauges planted on an area of land without any support may be subjected to turbulence, causing water to overflow out of the bucket.
36. (2pts.) True or false: satellite and RADAR both measure rainfall levels in the same manner. If false, explain why.
    Despite both instruments being suspended in the air, satellites measure rainfall levels from above, going downward, while RADAR emits wavelengths intended to shoot downward before projecting upward to be measured.
37. (3pts.) Given what you know about eutrophic and oligotrophic lakes, what can you infer about the environment of a mesotrophic lake?
    Since mesotrophic lakes are between eutrophic and oligotrophic lakes respectively, I can infer that there is some algae present, although less than the eutrophic portion, and that there is a reasonable amount of fish and partially clear, though not as clear as the oligotrophic lake.
38. (1pts.) True/False: All sinkholes are Karst topographic features
39. (2pts.) If you marked question 8 as false, justify your answer.
    Karst topographic features are defined as features formed when water dissolves on bedrock of any kind. Some sinkholes do form when water dissolves limestone, but there are other mechanisms that cause the formation of sinkholes that causes land to collapse. Only the sinkholes formed by water dissolving limestone are classified as Karst features.
40. (2pts.) Are ocean basins considered watersheds? Justify your answer.
    Ocean basins are not watersheds because only one tributary can feed into one watershed. Ocean basins are fed by many rivers, so they are not watersheds.
41. (2pts.) When is it considered for a stream to reach a dynamic equilibrium?
    The rate of deposition is equal to the rate of erosion, therefore a balance in water, energy, sediment, and vegetation as well.
42. (3pts.) What is channel degradation? What mechanism is it caused by? How does human activity contribute to this problem?

Channel degradation is when there is a decrease in runoff, less sediments, increase in streamflow, and a narrowing/steeping channel. This is caused by scouring, which is when a human-made obstruction causes sediment to build up at a local area as the velocity of water can no longer carry the sediments. Dams are an example of human activity contributing to channel degradation.

43. (1pt.) What type of pollution might you expect to be in reservoirs or anywhere with water?
    Non-point source pollution
44. (2pts.) As a hydrologist, you observe a local river and discover that its channel length is 57 meters long, and has a sinuosity of 2.1. What distance does this channel actually travel in straight length? What type of channel would this be considered? Round to the nearest hundredth.
    27.14 meters, considered sinuous or meandering, depending on the type of classification you use.
45. (5pts.) Which hypothesis best supports the rationale of the overland flow mechanism? Explain what the hypothesis proposes. **(Tiebreaker #4)**
    There is technically no single hypothesis that fully accurately accounts for all aspects of overland flow globally. The best hypotheses include Horton's infiltration excess overland flow hypothesis and Hewlett and Hewbert's saturated overland flow hypothesis and the variable source concept. Horton's hypothesis is relatively simple, stating that overland flow occurs when the rainfall/precipitation exceeds the soil infiltration rate. However, the simplicity of this hypothesis is not applicable in some scenarios. Hewlett and Hewbert's saturated overland flow hypothesis suggests that because the saturated zone is an extension of the river channel (variable source concept), all rainfall infiltrates the soil, and the water table rises above ground, appearing as overland flow.
46. (3pts.) Karst features are formed by which type of weathering? Explain how karst features in general form.
    Chemical weathering; when there is an abundance of carbon dioxide and water, it would undergo carbonation, forming carbonic acid that drips on limestone. The carbonic acid would then dissolve the limestone.
47. (2pts.) Provide an example of how human activity can be an accelerated form of chemical weathering.
    Increased nitrous and sulfur oxide emissions by industrial entities accelerates chemical weathering because these oxides can combine with water in the atmosphere to form nitric acid and sulfuric acid, respectively, which can accelerate weathering and be detrimental to the environment.
48. (1pt.) True/False: There is one universal stream order numbering system.
49. (1pt.) True/False: Honey, like salt, can also cause physical weathering by haloclasty.
50. (1pt.) True/False: Bedload is neither suspended nor bedded.

51. (3pts.) If a river's velocity increases as well as its turbidity, what would you expect the water temperature and photosynthesis to be (in other words, would it increase, decrease, or stay the same). If there is excess sediment (increased turbidity), how might it be detrimental to the environment?
    Increase in temperature (because of increased solar absorption), and decrease in photosynthesis because the organisms below the suspended sediments cannot absorb sunlight as well anymore. There is poor water quality, in some cases possibly eutrophication and destruction of the ecosystem.
52. (1pt.) What three mechanisms are primarily behind erosion?
    Wind, water, and ice
53. (7pts.) You have a relatively slow moving river with diverse sediment types. List the six sediment types and explain where you might find each type. Justify your answer.
    Boulder, cobble, clay, silt, pebble/granule, and sand. If the river is extremely slow and calm, visible sediments like boulder, cobble, and pebbles would likely be at the bottom, as bed load, while the less visible sediments: clay, silt, and sand would be suspended within the water column. This is because the river velocity in this scenario is not enough to transport larger sediment types.
54. (2pts.) You and your neighbor enter a well drilling competition on your respective lands. After drilling ninety meters in an attempt to allocate water, you lose miserably to your neither. Your neighbor, on the other hand, only drilled thirty meters before finding water, and casually spent the next few hours eating ice cream watching you dig. Is there a reason why your neighbor was able to dig a hole successfully while you didn't, despite the close proximity of the land, or was it just your bad luck? Explain your answer.
    Your neighbor's land likely sits atop a perched water table, which allowed some water to rise toward the well they dug, despite only digging thirty meters. Meanwhile, your land likely was above an impermeable layer/aquitard that had no water at all.
55. (2pts.) Describe the association between permafrost and pingos. What country has the most pingos?
    The development of permafrost around the edges of a dried up lake gives rise to pingos as water from the lake is pushed toward the center. Pingos often form in previous lakes that were surrounded by permafrost. Canada has the most pingos.
56. (6pts.) Describe the steps to the formation of a pingo. Number each step. There is no requirement for the number of steps, but be sure to be specific.
    a. 1. A lake surrounded by permafrost partially dries up.
    b. 2. Permafrost forms around the edges of the lake and on the lakebed, expanding the water and lake bed pushing the remaining water and sediment toward the center of the lake.
    c. 3. Water and sediment builds up in the center of the lake, and ultimately becomes permafrost.
    d. 4. Core of ice (permafrost) in a hill-like manner is a permafrost.

57. (12pts.) You are in a passionate debate with an adult who believes steps taken to halt climate change and melting permafrost are not necessary measures to be implemented. The adult claims that melting permafrost is beneficial for the environment since the melting ice will add more water to rivers and warm the local climate, so the human population should accelerate climate change. Provide a counterclaim centered on the effects of melting permafrost in relation to global warming. Be sure to describe at least five effects of melting permafrost. **(Tiebreaker #1)**
Sample answer: Like glacial melting, the rise in greenhouse gas emission has caused the global average temperature to increase, almost by a little every year. The slight changes to the average temperature have allowed permafrost all over the world to start melting. Permafrost contains many additional greenhouse gases, especially carbon, so once it melts, more carbon and even methane will be emitted to the atmosphere, creating this whole endless feedback loop to global warming. (1) Additionally, because permafrost has a lot of carbon and methane, this can seep into rivers and lakes, contaminating human drinking water. (2) From what we all know, mercury is an extremely toxic substance that can kill humans. Some permafrost contains high levels of mercury; as a result, once permafrost begins to melt, large amounts of mercury will be released into the atmosphere, contaminating lakes and rivers and perhaps even cause human casualties. (3) Given that we are in the midst of a pandemic, this reason is likely the most important: thousands or perhaps even millions of years ago, permafrost formed around microbes and pathogens, essentially "pausing" their ability to harm. Once permafrost melts, this may reactivate many of these pathogens and zombie bacteria that can cause human outbreaks. One example of this is the 2016 anthrax outbreak which was traced back to a deceased reindeer from decades prior that was exposed to microbes in permafrost. (4) Moreover, permafrost has upended the infrastructure of northern countries like Russia and Siberia, near the poles, damaging roads on a seasonal basis. This is because ice expands when it freezes, already initially damaging the roads (cracks, etc.) and once it melts, it further damages roads, causing it to crumble, etc. Therefore, from a political standpoint, permafrost has the potential to damage countries and states like Alaska. (5)
58. (3 pts.) Define solifluction. Why does solifluction only occur in the summer?
Solifluction is the creation of many lobes/wave-like features on permafrost when the uppermost layer of permafrost (active layer) melts and slides along the permafrost before eventually refreezing. This occurs only in the summer because the climate must be warm enough for the active layer of the permafrost to melt.
59. (2pts.) List at least one reason why dams may be disruptive.
Sample answer: Dams can alter water flow and sediments, which can cause a stream to move further away from dynamic equilibrium, leading to amplified sediment aggradation or degradation.

60. (2 pts.)What are the two subgroups of static karsts?

Zonal and Azonal karsts.

61. (3 pts.) You are following a stream when you find that it almost disappears into the land. How is this possible? What are some likely characteristics of the soil? How could you still trace the stream?
Sample answer: the land is likely riddled with karst features, meaning the soil is likely basic from a limestone composition. You could trace the stream using special biodegradable chemicals.

62. (3pts.) Imagine you are a scientist who proposes that we can acquire our energy through the energy rushing in streams. Given that a stream in dynamic equilibrium has no net input of energy, how can you alter the stream so that there is more energy output than input that we can use as a renewable energy source? List 3 ways.
Channelization, urbanization, decreased vegetation, dams, etc.

63. (3 pts.) Describe some practical applications of classification via stream orders.
Stream order allows for unique classification of streams and tributaries, which can be used to understand and manage the streams. Additionally, it can be used to model and track things like organisms and sediments.

64. (2pts.) Name the feature on the image below. This feature is a defining characteristic of what aspect of stream channels?



This is an example of an eyot. The presence of many eyots indicates that the channel it is surrounded by is a braided stream channel since braided channels are separated by eyots.

65. (2pts.) After being on TikTok for hours on end, you decide to leave your house for the first time in weeks and touch some grass. Because your house is a few miles away from a river, you notice a thin, even layer of soil being transported by runoff. Taking this into account as well as the image below, is erosion, deposition, or weathering currently occurring? If applicable, what type?

Erosion; gully/sheet erosion

66. (4pts.) List the association between flow rate, turbidity, water temperature, and photosynthesis rate.
    When flow rate increases, turbidity also increases, water temperature increases, but photosynthesis rate decreases.
67. (2pts.) How are clastic rocks classified? Identify the name of the classification.
    By size; Udden-Wentworth grain size scale
68. (2pts.) What function can contaminants have other than polluting groundwater?
    Measuring contaminant levels can determine the age of the groundwater by calculating its residence time.
69. (5pts.) List the 5 stages in the formation of a meander.
    1. Straight channels deepen along its bars
    2. The river oscillates, undercutting the outer bank and deposition starts to occur on the inside, but the channel itself is not widening yet.
    3. Hydraulic action & abrasion deepen the channel, and point bars form where deposition is occurring (inner bank)
    4. Helicoidal flow emerges, creating bends to the developing meanders
    5. Erosion progresses from the middle to the bend of a meander, and meanders move downstream
70. (2pts.) Wastewater derived from household drainages are called- (provide an example as well)
    Gray water; water used when showering goes down the drain and exits the house via pipes; this water is considered gray water.
71. (1pt.) Identify this feature:



    Tower karsts
72. (3pts.) Why is calculating the Darcy flux more favorable than a regular flux? **(Tiebreaker #2)**
    Flux alone assumes too many variables to be in a constant state, which is clearly not the case in groundwater. For example, flux does not take water tortuosity in groundwater into

account and does assume that water flows in a straight line underground. As a result, this can underestimate the actual flux. Darcy flux, on the other hand, accurately models groundwater flux as it takes porosity into account, which also indirectly implies water tortuosity.